

STAT 2507H Intro to Statistical Modeling I

Lecture Notes

Review lecture –

Basic concepts – I

- ▶ Population, sample, variables
- ▶ Type of data, type of variables (qualitative; quantitative (continuous, discrete))

Graphs for visualizing data

- ▶ Bar/pie chart
- ▶ dot plot, stem-and-leaf plot
- ▶ Shape of distributions (unimodal, bimodal)
- ▶ line chart (time series)

Describe quantitative data

- ▶ Mean, median, mode, range ($x_{max} - x_{min}$)
- ▶ (Populaton/sample) mean/variance/standard deviation
- ▶ Tchebysheff's theorem: At least this percentage

$$100(1 - 1/k^2)$$

of all measurements will lie in $[\mu - k\sigma, \mu + k\sigma]$

- ▶ Percentile, quartiles (the first, the second, the third), IQR, and box-plot (outlier detection)

Describe qualitative data

- ▶ side-by-side pie chart, stacked bar chart, bar chart with adjacent bars
- ▶ Scatterplots

Variables

- ▶ Dependent variables, independent variables
- ▶ Covariance
- ▶ Correlation coefficient (classifications: perfect positive/negative relationship; strong/moderate/weak, positive/negative linear relationship)
- ▶ The Least-squares regression line

Probability

- ▶ Outcomes, events, probability of an event
- ▶ How to assign probability
 - ▶ Method 1. relative frequency
 - ▶ Method 2. classical method – the sample space is equiprobable
 - ▶ Counting rules: the fundamental counting principle (i.e., using a product); permutations, combinations
- ▶ The law of large numbers
- ▶ Venn diagram to illustrate events and their operations; useful formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Probability and its properties

- ▶ Conditional probability
- ▶ The multiplication rule
- ▶ Independent events
- ▶ The law of total probability, Bayes' rule
- ▶ Probability tree

Random variables

- ▶ Discrete random variables
 - ▶ distribution (as a sequence of non-negative values with sum equal to 1)
 - ▶ mean, variance and standard deviation;

$$\sigma_X^2 = V(x) = E(X^2) - [E(X)]^2.$$

Special distributions

- ▶ Binomial
- ▶ Hypergeometric
- ▶ Poisson
- ▶ Approximation under suitable conditions – Hypergeometric approximated by binomial; Binomial approximated by Poisson

Continuous random variables

- ▶ Probability density function
- ▶ Cumulative probability $P(X \leq x)$
- ▶ Special distributions
 - ▶ Uniform on $[a, b]$; exponential
 - ▶ Their mean, variance/standard deviation
 - ▶ Normal, standard normal
 - ▶ For $X \sim N(\mu, \sigma^2)$ and $P(X \leq x)$, find the z-score and use the standard normal table to evaluate
 - ▶ Normal approximation of the binomial distribution (by matching the mean and variance)

Sampling plans

- ▶ Simple random sampling
- ▶ Other sampling plans
 - ▶ Stratified random sampling
 - ▶ Cluster sampling
 - ▶ 1-in-k systematic sampling
 - ▶ Biasedness and other potential problems after random selection

Statistics and sampling distributions

- ▶ Sampling distribution of the sample mean
 - ▶ It is still normal if the population distribution has a normal distribution
 - ▶ Central limit theorem for the general case ($n \geq 30$)
- ▶ CLT for the sample proportion:

$$N(p, pq/n)$$

if $np \geq 5$ and $nq \geq 5$.

Statistical inference

- ▶ Confidence interval for population mean
- ▶ Confidence level $(1 - \alpha)100\%$ with known σ

$$(\bar{X} - E, \bar{X} + E)$$

with margin of error

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- ▶ Need a normally distributed population
- ▶ Or non-normally distributed population with $n \geq 30$
- ▶ CI with σ unknown

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

- ▶ Need normally distributed population
- ▶ If $n - 1 \geq 30$, we approximate $t_{\alpha/2, n-1}$ by $z_{\alpha/2}$.

- ▶ CI for the population proportion

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

We need $n\hat{p} \geq 5$, $n\hat{q} \geq 5$.

- ▶ The use of p^* (see Week 9, lesson 1)

Compare two population means

- ▶ σ_1 and σ_2 are known: CI for $\mu_1 - \mu_2$

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- ▶ σ_1 and σ_2 unknown
 - ▶ $\sigma_1 \neq \sigma_2$; use $df = \min(n_1 - 1, n_2 - 1)$
 - ▶ $\sigma_1 = \sigma_2$; use pooled sample variance s_p^2 (see Week 10, lesson 1)
 - ▶ How to decide? Rule of thumb:

$$1/3 \leq s_1^2/s_2^2 \leq 3$$

- ▶ CI will use the t -distribution

Compare two population proportions

- For the CI, we use

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

Hypothesis tests

- ▶ H_0 (status quo position) and H_a
- ▶ the p -value (as a probability), calculated assuming H_0
- ▶ significant level α
- ▶ The rule:
 - ▶ If $p\text{-value} \leq \alpha$, reject H_0 and accept H_a
 - ▶ If $p\text{-value} > \alpha$, we fail to reject H_0

Hypothesis test for a population mean μ

- ▶ The three types of tests for μ
 - ▶ Right-tailed test
 - ▶ Left-tailed test
 - ▶ Two-tailed test
- ▶ σ known, the p -value is $P(Z > z_0)$ ($H_a: \mu > \mu_0$).
- ▶ σ unknown, use (t_0, T_{n-1}) in place of (z_0, Z) .

Hypothesis test for a population proportion p

- ▶ We only use the statistic

$$z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$$

- ▶ No t -distribution is used.

The critical value method for the population mean or proportion

- ▶ Compare z_0 with $z_{\alpha/2}$
- ▶ Compare t_0 with $t_{\alpha/2}$ under unknown population variance (need normal population distribution)

Type I and II errors

- ▶ Type I: H_0 is correct but we reject it.
- ▶ Type II: H_0 is false but we fail to reject.

How to remember: Type I error is more serious (think of an innocent person being convicted)

$$P(\text{Type II error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false}) = \beta$$

Power is test is $1 - \beta$

Hypothesis tests for comparing two population means

- ▶ Right-tailed test. $H_0: \mu_1 - \mu_2 = D_0$. $H_a: \mu_1 - \mu_2 > D_0$.
- ▶ Left-tailed test. $H_0: \mu_1 - \mu_2 = D_0$. $H_a: \mu_1 - \mu_2 < D_0$.
- ▶ Two-tailed test. $H_0: \mu_1 - \mu_2 = D_0$. $H_a: \mu_1 - \mu_2 \neq D_0$.

Method:

- ▶ If σ_1 and σ_2 are known, compute the p -value using the normal distribution
- ▶ If σ_1 and σ_2 are unknown, compute the p -value using the t -distribution

(Week 12, lesson 2)

Hypothesis tests for comparing two population proportions

Compute the p -value using the normal distribution

Statistical inference for paired-differences

$$(x_i, y_i) \longrightarrow d_i = x_i - y_i$$

- ▶ Confidence interval for μ_d
- ▶ Hypothesis test for μ_d .
 - ▶ Compute the p -value using the t -distribution if σ_d is unknown.
 - ▶ (Compute the p -value using the normal distribution if σ_d is given)